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The selection of popular trading strategies

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Abstract

Purpose – The relative performance of five popular nonzero-investment strategies, including Size, book-to-market ratios, earnings-to-price (E/P) ratios, cash flow-to-price (CF/P) ratios and dividend-to-price ratios, and their corresponding zero-investment strategies (also known as premiums) are first examined altogether for equally weighted (EW) and value-weighted (VW) methods to check whether a certain strategy (or some strategies) could be recommended to portfolio managers as the best (better) strategy (strategies). The paper aims to discuss these issues.

Design/methodology/approach – This paper uses the stochastic dominance (SD) approach, a non-parametric test, to investigate the relative performance among various strategies and help investors search for the best or better strategy (strategies).

Findings – The main results show that both the highest E/P and CF/P strategies (and their corresponding premiums) generally produce higher returns than the other three strategies (and their corresponding premiums) through allocating investors' capital between the risky and risk-free assets for the EW and VW methods, respectively.

Research limitations/implications – This study only examines US stock markets by SD approach, whether the results are consistent with non-US markets still needs further investigation. The findings imply that investors can benefit by investing in the highest E/P or CF/P stocks (or their corresponding premiums) to make more profit or less loss for US stock markets.

Practical implications – First, the SD findings suggest that investors or portfolio managers can allocate their funds between risky and risk-free assets to maximize their profits. Next, the simulation results again prove that the profits of each nonzero-investment or zero-investment strategy for EW portfolios are higher than those of each corresponding strategy for VW portfolios. Finally, the findings imply that portfolio managers or investors can invest in the highest E/P or CF/P stocks (or their corresponding premiums) to make more profit or less loss.

Originality/value – This study first uses an extensive data set (1952-2009) to examine the relative performance of nonzero-investment strategies and their corresponding zero-investment strategies for the five popular indicators altogether for the EW and VW methods with the SD approach for US stock markets. Moreover, the results reveal that the investors or portfolio managers can invest in the highest E/P and/or CF/P portfolios (or their corresponding premiums) to make more profit or less loss.

Keywords Investment analysis, Investments

Paper type Research paper

1. Introduction

Many empirical studies have documented stock market anomalies[1] and dubbed them as nonzero-investment (nonzero-cost) strategies[2]. Among these, fundamental variables are often related to stock returns.

The most prominent firm anomaly is the so-called size effect, which asserts that smaller firms have higher average risk-adjusted returns than larger firms. Many studies, such as Banz (1981), Reinganum (1981), Keim (1983), Stoll and Whaley (1983),



Fama and French (1992), Lau *et al.* (2002), and Cenesizoglu (2011), find the size effect in US and non-US stock markets. However, Eraslan's (2013) recent study shows that the big and medium size portfolios overwhelm small size portfolios on realized excess returns in the Istanbul stock market.

The accounting anomalies are defined as changes in stock prices that occur after the release of accounting information. The most famous accounting anomaly is that returns from firms with higher earnings-to-price (E/P) ratios outperform those with lower E/P ratios (Basu, 1977; Goodman and Peavy, 1983; Levy and Lerman, 1985; Chou and Liao, 1996; Lau *et al.*, 2002; Kyriazis and Christou, 2013).

Some studies also point out that returns on higher book-to-market equity (B/M) portfolios are higher than those on lower B/M portfolios in US, Japanese and UK markets (Basu, 1983; Chan *et al.*, 1993; Fama and French, 1992, 2008; Lakonishok *et al.*, 1994; Lau *et al.*, 2002; Cenesizoglu, 2011; Eraslan, 2013; Kyriazis and Christou, 2013). In addition, Ko *et al.* (2014) have found that a zero-investment portfolio, constructed by buying the highest B/M and short-selling the lowest B/M portfolios based on the moving average rule, can produce significantly positive returns.

Chan *et al.* (1993), Davies (1994), Lau *et al.* (2002) and Hou *et al.* (2011) also report that firms with higher cash flow-to-price (CF/P) ratios demonstrate better performance than those with lower CF/P ratios in US and some non-US stock markets. Besides, the findings of Gharghori *et al.* (2013) reveal that high CF/P ratios portfolios dominate low CF/P ratios portfolios in the Australian stock markets. Fama and French (1988) and Kyriazis and Christou (2013) document another accounting anomaly, dividend-to-price (D/P) effect. They find that returns from firms with higher D/P ratios outperform those with lower D/P ratios.

Comparing the performance among these investment strategies is an important and interesting issue for practitioners and academicians. Senchack and Martin (1987), an early study, document that the high E/P strategy dominates the high sales-to-price ratio strategy on both an absolute and risk-adjusted basis in US stock markets. Hart *et al.* (2003) examine the profitability of a broad range of stock selection strategies, including the Size, B/M, E/P and D/P indicators, for 32 emerging markets. Their results show that the high E/P strategy generates the higher average excess returns than all the other strategies. Park and Lee (2003) also compare the similar strategies for the Japanese stock market and find that no unique strategy can outperform all the other strategies across the industries[3].

Our study is important and interesting for four reasons. First, Senchack and Martin, as noted above, only examine the relative performance of two investment strategies for US markets. The Size, B/M, E/P, CF/P and D/P portfolios, collected and synthesized by the Kenneth R. French-Data Library, are probably the most important and popular indicators in US stock markets. To the best of our knowledge, however, there is little empirical study examining the relative performance of these five strategies altogether in the literature[4]. Therefore, the relative returns among these five strategies, the nonzero-investment strategies, are first investigated altogether to check whether a certain strategy (or some strategies) could be recommended to portfolio managers as the best (better) strategy (strategies).

Second, as mentioned above, the relative performance of stock selection strategies is often measured as the returns on a nonzero-investment strategy, making an investment with the investors' own capital, as in the work of Senchack and Martin (1987) and Chou and Liao (1996). The relative performance of investment strategies can also be measured as the returns on a zero-investment strategy (also known as a premium),

involving a long position in the winner portfolio and an offsetting short position in the loser portfolio, as in the studies of Hart *et al.* (2003) and Park and Lee (2003). The investors (or speculators) may adopt a nonzero-investment strategy to make an investment with their own money while arbitragers may adopt a zero-investment strategy to make an investment without tying up any of their own capital. For further study, the relative performance of the five zero-investment (premium) strategies is also first examined altogether in this study.

Third, the weighting of individual stocks within the portfolio is an important factor to the performance evaluation. The weight that a portfolio manager assigns to a given stock in a portfolio can make a contribution to return that is as important as the investment timing and stock selection decisions. Generally speaking, the equal-weighted (EW) and value-weighted (VW) methods are two popular mechanical schemes for portfolio weighting[5]. Pae and Sabbaghi (2010) prove that EW portfolios (method) outperform VW portfolios (method)[6]. Recently, Plyakha *et al.* (2012) also demonstrate that EW portfolios dominate not only VW but also price-weighted portfolios. Senchack and Martin (1987), Hart *et al.* (2003) and Park and Lee (2003), as mentioned above, only examine the relative performance among various strategies for EW portfolios. To examine how two different weighting rules affect the selection of trading strategies, the relative performance of each strategy for the EW and VW methods is first investigated in this study.

Fourth, with regard to the research methodology, the firm or accounting anomalies tests are almost carried out within the famous Capital Asset Pricing Model (CAPM) or market model in the literature[7]. However, Roll (1977) seriously doubts the validity of the CAPM on both empirical and theoretical grounds; therefore, various anomalies may be interpreted as rejecting the CAPM hypothesis rather than efficient market hypothesis[8]. The disadvantages of the CAPM and other selection criteria are that they almost rely on the assumption of a normal distribution of returns and/or the concave investor utility function. The normality assumption of stock returns is obviously inappropriate because Fama (1965) and Schwert (1990), for example, have documented that stock return distributions are fat tailed, that is, the extreme tails of return distributions carry higher probability density than normal distribution.

To avoid the argumentations in previous literature, this study first uses stochastic dominance (SD) approach, a non-parametric test, to investigate the relative performance of the five nonzero-investment and zero-investment strategies for EW and VW methods for US stock markets, respectively, and intends to help investors search for the best (better) strategy (strategies). The first advantage of the SD analysis over parametric tests becomes apparent when the stock return distribution is non-normal, as the SD approach does not require any assumption about the nature of the distribution and therefore can be used for any type of distribution. Next, the SD theory, revealing the entire distribution, recovers all information from the distribution while traditional parametric tests, depending only on the mean and variance, omit all information from higher moments. Third, the SD theory makes less restrictive assumptions regarding investor utility functions while the CAPM model is derived based on the assumption that investor utility functions must be concave.

Finally, the SD allows some proportion of investors' money to be invested in a risky asset and the remaining to be invested in a risk-free asset. Our methodology equips investors with a better tool for asset allocation. This approach is superior and less restrictive than the traditional parametric statistics for making investment decisions under uncertainty[9].

Our main results show that both the highest E/P and CF/P strategies (and their corresponding premiums) generally produce higher returns than the other three strategies (and their corresponding premiums) through allocating investors' capital between the risky and risk-free assets in US and non-US financial markets for the EW and VW methods, respectively[10]. Furthermore, the same or similar phenomena are found in each sub-sample period and different holding periods. These findings imply that portfolio managers can benefit by investing in the highest E/P and/or CF/P stocks (or their corresponding premiums).

The remainder of this paper is organized as follows: The second section describes the sample data and methodology. The third section presents the empirical results. Robustness tests and simulations are examined in the fourth section. The fifth section gives significance tests and practical implications, and the conclusions are presented in the last section.

2. Sample data and methodology

2.1 Sample data

This study uses the traditional design to search for a nonzero-investment strategy for each indicator. Our sample firms are listed on the NYSE or AMEX or NASDAQ. The five indicator portfolios examined comprise stocks with similar values of Size and similar ratios of B/M, E/P, CF/P and D/P for a one-year holding period, respectively.

Nonzero-investment portfolios of five indicators are formed every year, respectively. For example, 57 time series of market values (Size) are obtained for each of the sample firms, corresponding to the 57 years from 1952 through 2009. The annual size cross-sections are sorted based on the ascending markets values, and ten portfolios are formed at the end of June of every year according to the market values at the end of the previous year[11]. The Size 1 portfolio consists of firms with the lowest market values and each subsequent portfolio contains the firms with the next lowest market values, and accordingly the Size 10 portfolio consists of the firms with the highest market values. The other four indicators, the B/M, E/P, CF/P and D/P ratios, also use the same method to form and rebalance their portfolios each year. Therefore, each of the portfolios related to the Size, B/M, E/P, CF/P and D/P indicators has 57 time series of data, corresponding to the sample period. The annual returns (from July of t year to June of $t + 1$ year) on each portfolio of five indicators are calculated by compounding the monthly returns obtained from the Kenneth R. French Data Library over a 57-year period of 1952-2009[12]. We use the SD approach to select the winner portfolio, dubbed it as a strategy, for each indicator.

After the selection of the five nonzero-investment strategies, the corresponding five zero-investment strategies, involving a long position in the winner portfolio and an offsetting short position in the loser portfolio, are also constructed every year. As in previous studies, these strategies are also dubbed as the Size, B/M, E/P, CF/P and D/P premiums.

Panels (a) and (b) of Table I present the average annual returns and standard deviations of ten portfolios of each indicator for the EW and VW methods. The highest returns on the size indicator (18.32 percent in panel (a) and 15.48 percent in panel (b)) occur in the smallest portfolios (Size 1) for the EW and VW methods, which seem to show the existence of the size effect. The B/M, E/P and CF/P effects also seem to exist as the highest returns occur in their highest ratio portfolios, no matter in panels (a) or (b). However, the D/P effect is not so clear because the highest returns occur in the eighth portfolio. The figures of the table also present that the winner of each indicator for the

Panel (a): five investment indicators for the EW method ^a										
Portfolio	Size		B/M		E/P		CF/P		D/P	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
1 (Smallest)	0.1832	0.3009	0.0926	0.2878	0.1232	0.2802	0.1158	0.2756	0.1488	0.2675
2	0.1406	0.2525	0.1227	0.2445	0.1368	0.2363	0.1319	0.2328	0.1585	0.2234
3	0.1431	0.2298	0.1379	0.2294	0.1411	0.2253	0.1454	0.2325	0.1563	0.2329
4	0.1381	0.2318	0.1542	0.2368	0.1476	0.2116	0.1502	0.2136	0.1616	0.2167
5	0.1396	0.2227	0.1594	0.2264	0.1566	0.2166	0.1647	0.2096	0.1637	0.2115
6	0.1351	0.2102	0.1720	0.2296	0.1652	0.2054	0.1695	0.2055	0.1672	0.2147
7	0.1353	0.2054	0.1799	0.2244	0.1739	0.2154	0.1851	0.2163	0.1699	0.2084
8	0.1261	0.1932	0.1834	0.2218	0.1830	0.2113	0.1838	0.2122	0.1729	0.2043
9	0.1239	0.1802	0.2063	0.2354	0.2003	0.2289	0.2037	0.2336	0.1596	0.1917
10 (Largest)	0.1079	0.1775	0.2257	0.2891	0.2209	0.2583	0.2269	0.2647	0.1410	0.1852
Panel (b): five investment indicators for the VW method ^b										
1 (Smallest)	0.1548	0.2761	0.1088	0.2206	0.1103	0.2476	0.1072	0.2279	0.1230	0.2469
2	0.1459	0.2546	0.1148	0.1860	0.0972	0.1790	0.1127	0.1929	0.1118	0.1920
3	0.1480	0.2305	0.1202	0.1826	0.1160	0.1816	0.1079	0.1868	0.1178	0.1823
4	0.1430	0.2326	0.1173	0.1863	0.1134	0.1764	0.1202	0.1851	0.1238	0.1838
5	0.1442	0.2205	0.1287	0.1758	0.1205	0.1739	0.1294	0.1902	0.1135	0.1866
6	0.1374	0.2101	0.1307	0.1846	0.1417	0.1785	0.1239	0.1858	0.1263	0.1726
7	0.1377	0.2057	0.1294	0.1848	0.1491	0.1888	0.1308	0.1799	0.1271	0.1882
8	0.1297	0.1908	0.1510	0.2030	0.1540	0.1960	0.1404	0.1791	0.1432	0.1786
9	0.1256	0.1767	0.1547	0.1897	0.1690	0.1960	0.1638	0.1938	0.1356	0.1784
10 (Largest)	0.1097	0.1761	0.1640	0.2364	0.1839	0.2372	0.1788	0.2240	0.1212	0.1941
Panel (c): five zero-investment strategies (premiums)										
Strategy portfolio	Size premium		B/M premium		E/P premium		CF/P premium		D/P premium	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Equally-weighted	0.0753	0.2363	0.1332	0.1836	0.0977	0.1478	0.1110	0.1478	0.0240	0.1334
Value-weighted	0.0452	0.2203	0.0552	0.1880	0.0736	0.1945	0.0716	0.1875	0.0203	0.1758

Notes: ^aThe means and standard deviations of the annual returns on the EW index, taken from the CRSP Database, are 15.18 and 23.82 percent during the sample period; ^bthe means and standard deviations of the annual returns on the VW index, taken from the CRSP Database, are 11.55 and 17.89 percent during the sample period

Notes: ^aThe means and standard deviations of the annual returns on the EW index, taken from the CRSP Database, are 15.18 and 23.82 percent during the sample period; ^bthe means and standard deviations of the annual returns on the VW index, taken from the CRSP Database, are 11.55 and 17.89 percent during the sample period

Table I.
Means and standard
deviations of annual
returns on various
indicators and
premiums during the
1952-2009 period

EW method outperform its corresponding indicator for the VW method. For example, the average annual returns on the B/M 10 portfolio for the EW method (22.57 percent) are higher than those on the B/M 10 portfolio for the VW method (16.40 percent), and the other four indicators also present the same phenomena. In addition, the average annual returns on the Size and D/P premiums, as shown in panel (c), are lower than those on the other three premiums for the EW and VW methods. The average annual returns on each premium for the EW method also dominate those on its corresponding premium for the VW method.

2.2 Methodology[13]

SD rules provide optimal investment criteria under uncertainty for all expected utility maximizers with given preference structures. Thus let U_1 be the class of all non-decreasing utility functions ($u' > 0$ for every u in U_1); U_2 , the class of all concave non-decreasing utility functions ($u' > 0$ and $u'' < 0$ for every u in U_2). Also, let F and G be the cumulative probability distributions of two uncertain options with quantiles $Q_F(p)$ and $Q_G(p)$ respectively.

The following SD rules apply to all investors with utility functions in class U_1 and U_2 , respectively, when only risky investment options are available and there is no risk-free asset (no borrowing and lending at the risk-free rate is permitted). First degree stochastic dominance (FSD) and second degree stochastic dominance (SSD) are first introduced.

FSD: option F dominates option G for all investors with utility functions in U_1 if and only if: $Q_F(p) \leq Q_G(p)$ for all p (with strict inequality for at least one p).

SSD: option F dominates option G for all investors with utility functions in U_2 if and only if: $\int_0^p Q_F(t)dt \leq \int_0^p Q_G(t)dt$ for all p (with strict inequality for at least one p).

When borrowing and lending at the risk-free rate is allowed, we can use much stronger rules, called SDR (SD with risk-free asset) rules. By analogy to FSD and SSD, these rules are denoted by first degree stochastic dominance with risk-free asset (FSDR) and second degree stochastic dominance with risk-free asset (SSDR) for investors with preference structures representing by utility functions from U_1 and U_2 , respectively.

Now we define new random variables X_α and Y_β as follows: $X_\alpha = (1-\alpha)r_f + \alpha X$ and $Y_\beta = (1-\beta)r_f + \beta Y$, where α ($0 \leq \alpha \leq 1$) and β ($0 \leq \beta \leq 1$) are positive constants. F_α and G_β denote the cumulative distributions of X_α and Y_β , while $\{F_\alpha\}$ and $\{G_\beta\}$ denote the sets of distributions of all possible mixes of X and r_f as well as Y and r_f , respectively. $\{F_\alpha\}$ dominates $\{G_\beta\}$ if and only if for each element $G_\beta \in \{G_\beta\}$ there exists at least one element in $\{F_\alpha\}$ which dominates it. To simplify the expression in the following formula, we define:

$$\delta(p) \equiv [Q_G(p) - r_f] / [Q_F(p) - r_f] \quad (1)$$

$$\gamma(p) \equiv \int_0^p [Q_G(t) - r_f] dt / \int_0^p [Q_F(t) - r_f] dt \quad (2)$$

FSDR: option F_α dominates option G_β for all investors with utility functions in U_1 if and only if: $\inf_{0 < p < F(R)} \delta(p) \geq \sup_{F < p < 1} \delta(p)$.

SSDR: option F_α dominates option G_β for all investors with utility functions in U_2 if and only if: $\inf_{0 < p < p_0} \gamma(p) \geq \sup_{p_0 < p < 1} \gamma(p)$, where p_0 is determined by the equation $\int_0^{p_0} [Q_F(t) - r_f] dt = 0$.

We use both the SD and SDR criteria to test the dominant relationship among various portfolios or strategies.

3. Empirical results

We first use CDF's curves of the B/M portfolios to present the application of the SD rules for EW portfolios in the subsection 3.1. The selection of five nonzero-investment strategies is shown in the subsection 3.2. The relative performance among the five nonzero-investment strategies is compared in the subsection 3.3. Finally, the relative performance among various zero-investment strategies is examined.

3.1 The CDF's curves of the B/M portfolios

We present an example of the application of the SD rules in this subsection. Figure 1 shows the CDF's curves of the B/M 1 and B/M 10 portfolios for the EW method. Generally, the curve of the B/M 10 lies to the right of the curve of the B/M 1, so it is questionable whether the B/M 10 dominates the B/M 1. Yet, the CDF's curves cross each other several times within the return range of about -30.0 percent and -21.0 percent. This indicates that the B/M 10 does not dominate the B/M 1 according to the FSD rule, under which the only restriction on the risk preference structure is that the investor utility function is non-decreasing ($u' \geq 0$).

Yet, according to the SSD, the B/M 10 obviously dominates the B/M 1, because all of the areas between the vertical axis and the CDF of the B/M 10 are always greater than the area enclosed by the CDF of the B/M 1. Consequently, the cumulative area between the two distributions remains positive. In such a situation, all risk-averse investors would prefer the B/M 10 to B/M 1 according to the SSD rule. As SSD implies SSDR, the B/M 10 dominates the B/M 1 when risk-averse investors are permitted to mix their B/M 10 with a risk-free asset by borrowing or lending, regardless of the return on a risk-free asset.

In the next subsection, the selection of various nonzero-investment strategies with SD rules is discussed in detail.

3.2 The selection of five nonzero-investment strategies

We use the FSD, SSD, FSDR and SSDR criteria to select the strategy (winner portfolio) from ten size portfolios and the market, as a demonstration, for the EW method. Their efficient sets are shown in Table II. The main findings are as follows.

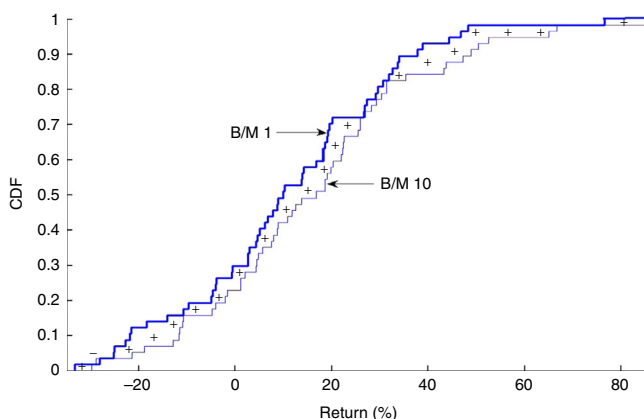


Figure 1.
The cumulative
return distributions
of the B/M 1 and B/
M 10 portfolios for
the EW method

MF
41,6

570

	FSD ^b	SSD	FSDR ^a 0.4% ≤ r_f ≤ 12.1%	1.8% ≤ r_f ≤ 2.4%	SSDR 4.9% ≤ r_f ≤ 6.0%	r_f ≥ 6.1%
Size 1 ^c	+ ^d	+	+	+	+	+
2	+	–	+	–	–	–
3	+	+	+	+	–	–
4	+	+	+	–	–	–
5	+	+	+	+	–	–
6	+	+	+	+	–	–
7	+	+	+	+	+	–
8	+	+	+	+	–	–
9	+	+	+	+	–	–
Size 10	+	+	+	+	–	–
Market	+	+	+	+	–	–

Notes: ^aAs noted in Footnote 13, the range of r_f between 0.4 and 6.0 percent, is generally used as the testing interval. ^bFSD, first-degree stochastic dominance; SSD, second-degree stochastic dominance; FSDR, first-degree stochastic dominance with a risk-free asset; SSDR, second-degree stochastic dominance with a risk-free asset. ^cSize 1, Size 10 and market denote the smallest-cap portfolio, the largest-cap portfolio and the equally-weighted market index, respectively. ^dThe efficient set (winners) is marked by “+” and the inefficient set (losers) is marked by “–”

Table II.

The selection of the Size strategy for the EW method

First, applying the weakest assumption ($u' \geq 0$) on the investors' preferences, we cannot distinguish among ten size portfolios and the market; i.e., all portfolios are included in the FSD (without borrowing or lending at a risk-free interest rate) efficient set. Under the FSDR rule (with borrowing or lending at a risk-free interest rate)[14], we find that all of the size portfolios and the market are still included in the FSDR efficient set. Thus, sharper decision rules are required to distinguish among various size portfolios.

Second, under the SSD rule, assuming the investors are risk-averse ($u' \geq 0$ and $u'' \leq 0$), the result reveals a slightly different phenomenon: the Size 2 (loser) is excluded from the SSD efficient set. The analytical results are stronger when the investors are permitted to borrow or lend money at a risk-free interest rate. The size of an SSDR efficient set is[15], of course, a decreasing function of the assumed risk-free interest rate. The Size 1 portfolio is the winner for $r_f \geq 6.1$ percent [16]. This indicates that the smallest-cap portfolio is selected as one of the five nonzero-investment strategies.

An important and interesting phenomenon appears in Table II. The larger size portfolios are gradually excluded from the SSDR efficient sets when the r_f gradually increases. Therefore, in the lower right-hand corner of the SSDR efficient sets there appears the “–” signs (losers). The evidence indicates that a certain portfolio can beat some portfolios through asset allocation, namely, some proportion of the investors' money is invested in a risky asset and the remaining in a risk-free asset.

Next, the other four nonzero-investment strategies are selected by the same procedure as well. The selecting results of each indicator for the EW method are summarized in Table III.

The table shows only the B/M 10 portfolio, shown in Columns 3 and 4, is included in the SSDR efficient set for $r_f \geq 5.2$ percent. That is, the largest B/M portfolio outperforms all of the lower B/M portfolios and the market. Besides, the E/P 10 portfolio dominates the lower E/P portfolios and the market for $r_f \geq 2.2$ percent and the CF/P 10 portfolio is

Size portfolios	SSDR ^a $r_f \geq 6.1\%$ ^b	B/M portfolios	SSDR $r_f \geq 5.2\%$	E/P portfolios	SSDR $r_f \geq 2.2\%$	CF/P portfolios	SSDR $r_f \geq 3.6\%$	D/P portfolios	SSDR $r_f \geq 5.1\%$
Size 1 ^c	^d +	B/M1	-	E/P1	-	CF/P1	-	D/P1	-
2	-	2	-	2	-	2	-	2	-
3	-	3	-	3	-	3	-	3	-
4	-	4	-	4	-	4	-	4	-
5	-	5	-	5	-	5	-	5	-
6	-	6	-	6	-	6	-	6	-
7	-	7	-	7	-	7	-	7	-
8	-	8	-	8	-	8	-	8	+
9	-	9	- ^e	9	- ^e	9	- ^e	9	-
Size 10	-	B/M10	+	E/P10	+	CF/P10	+	D/P10	-
Market	-	Market	-	Market	-	Market	-	Market	-

Notes: ^aSSDR, as noted above, denotes second-degree stochastic dominance with a risk-free asset. ^bAs noted in the Footnote 13, the range of r_f between 0.4 and 6.0 percent, is generally used as the testing interval. ^cSize 1, as noted above, denotes the smallest-cap portfolio and Size 10 denotes the largest-cap portfolio, and the meaning of various B/M, E/P, CF/P, D/P and Market portfolios are discussed in the Data and Methodology section. In order to save space, only the SSDR results of each indicator are shown in the table. ^dThe efficient set (winners) is marked by "+" and the inefficient set (losers) is marked by "-". ^eAfter gradually increasing the risk-free interest rate, slightly higher than 6.0 percent, the findings show that the B/M 10 beats the B/M 9, the E/P 10 dominates the E/P 9, and the CF/P 10 is superior to the CF/P 9

Table III.
The selection of five
strategies for the
EW method

superior to the lower CF/P portfolios and the market for $r_f \geq 3.6$ percent. Finally, the D/P 8 portfolio beats other D/P portfolios and the market, that is, the stocks with higher D/P ratios tend to outperform the other portfolios.

In short, the Size 1, B/M 10, E/P 10, CF/P 10 and D/P 8 portfolios are selected as the five nonzero-investment strategies for the EW method[17]. An interesting question is which trading strategy can earn the higher profits for the EW and VW methods, which is examined in the next subsection.

3.3 The relative performance among nonzero-investment strategies

The comparative results of the five nonzero-investment strategies, the Size 1, B/M 10, E/P 10, CF/P 10 and D/P 8 strategies, are shown in panels (a) and (b) of Table IV for the EW and VW methods. The following conclusions can be drawn from the table.

First, all strategies are included in their FSD efficient sets for the two methods. Two FSDR efficient sets also reveal the same findings. Thus, sharper decision rules are again used to distinguish among various strategies.

Second, assuming that the investors are risk-averse ($u' \geq 0$ and $u'' \leq 0$), namely, the SSD rule can be used to investigate the dominant relationship among the winner strategies. The results show that both the Size 1 and B/M 10 strategies are excluded from their SSD efficient sets for the two methods. The analytical results are slightly different when the investors are permitted to borrow or lend money at the risk-free interest rate. We find that the E/P 10 and CF/P 10 are the last winners for the two methods, respectively.

In short, the E/P 10 and CF/P 10 strategies are the winners among the five nonzero-investment strategies for the two methods, respectively. Our findings indicate

	FSD ^b	SSD	SSDR ^a 0.4% ≤ r_f ≤ 1.6%	SSDR ^a 1.7% ≤ r_f ≤ 6.0%
<i>Panel (a): equally-weighted method</i>				
Size 1	+ ^c	–	–	–
B/M 10	+	–	–	–
E/P 10	+	+	+	+
CF/P 10	+	+	+	+
D/P 8	+	+	+	–
	FSD ^b	SSD	SSDR ^a 0.4% ≤ r_f ≤ 2.2%	SSDR ^a 2.3% ≤ r_f ≤ 6.0%
<i>Panel (b): value-weighted method</i>				
Size 1	+ ^c	–	–	–
B/M 10	+	–	–	–
E/P 10	+	+	+	+
CF/P 10	+	+	+	+
D/P 8	+	+	+	–

Table IV.
The relative
performance
of the five
nonzero-investment
strategies

Notes: ^aAs noted in Footnote 13, the range of r_f , between 0.4 and 6.0 percent, is generally used as the testing interval. ^bFSD, first-degree stochastic dominance; SSD, second-degree stochastic dominance; FSDR, first-degree stochastic dominance with a risk-free asset; SSDR, second-degree stochastic dominance with a risk-free asset. The FSDR efficient sets are not shown in the table as the results are the same as the FSD efficient sets. ^cThe efficient set (winners) is marked by “+” and the inefficient set (losers) is marked by “–”

that investors can benefit by investing in the highest E/P or CF/P stocks through allocating investors' capital between the risky and risk-free assets in US and non-US financial markets for the two methods, even though the risk-free rates are in the age of low levels (see Footnote 16 in detail).

3.4 The relative performance among zero-investment strategies

Corresponding to the five nonzero-investment strategies, the comparison of the five zero-investment strategies (premiums) is examined in this subsection. For the two methods, the efficient sets for various strategies are shown in panels (a) and (b) of Table V. We can draw several conclusions from the table.

For the EW method, assuming that the investors are risk-averse ($u' \geq 0$ and $u'' \leq 0$), i.e., the SSD rule can be used to distinguish among various premiums. The results of panel (a) show that the Size and D/P premiums are excluded from the SSD efficient set. The SSDR efficient set also presents the same finding in the age of low levels of the risk-free interest rate for $0.4\% \leq r_f \leq 2.2\%$ [18].

For the VW method, assuming that the investors are risk-averse ($u' \geq 0$ and $u'' \leq 0$), that is, the SSD rule is used to distinguish among various premiums. The result indicates that the Size and D/P premiums are excluded from the SSD efficient set. Besides, as shown in panel (b), the SSDR efficient set only includes the E/P and CF/P premiums for $0.4\% \leq r_f \leq 6.0\%$.

In short, the intersection of the winner strategies of the EW and VW methods is the E/P and CF/P premiums for $0.4\% \leq r_f \leq 2.2\%$. These results also imply that the arbitragers can make more profits by implementing the E/P or CF/P premium strategy.

4. Robustness tests and simulations

The above findings are further examined in this section. First, we divide the whole sample period into two almost-equal sub-sample periods, 1952 July-1980 June and 1980 July-2009 June, to investigate whether the results present a structural change in the different sub-sample periods. Second, different holding periods, the one-quarter, two- and three-year periods, are also examined to verify whether the results of the one-year holding period mentioned above are robust. The findings are discussed as follows.

4.1 Two sub-sample periods

The findings in the two sub-sample periods for the EW method suggest that the Size 1, B/M 10, E/P 10, CF/P 10 and D/P 8 portfolios are still the winners of the five indicators [19]. As for the VW method, the results of the two sub-sample periods display almost the same phenomena except a slightly different for the B/M indicator during the 1952 July-1980 June period [20].

Next, the relative performance of the five nonzero-investment strategies for the two sub-sample periods is examined. For the EW method, the evidences show that the E/P 10 and CF/P 10 strategies are still the winners in the first sub-sample period and the B/M 10 and CF/P 10 strategies dominate the other three strategies in the second sub-sample period. As for the VW method, the E/P 10 and CF/P 10 strategies outperform the other three strategies for the two sub-sample periods.

The comparison of various zero-investment strategies for the EW method shows that the E/P and CF/P premiums are winners in the first sub-sample period and the B/M and CF/P premiums dominate the other three premiums in the second sub-sample period.

Table V.
The relative performance of the five zero-investment strategies

MF
41,6

	FSD ^b	SSD	0.4% ≤ r_f ≤ 2.2%	SSDR ^a 2.3% ≤ r_f ≤ 4.3%	r_f ≥ 4.4%
<i>Panel (a): equally-weighted method</i>					
Size premium	+	–	–	–	–
B/M premium	+	+	+	+	+
E/P premium	+	+	+	–	–
CF/P premium	+	+	+	+	–
D/P premium	–	–	–	–	–
				SSDR ^a	
			0.4% ≤ r_f ≤ 6.0%	r_f ≥ 6.5%	
		SSD			
<i>Panel (b): value-weighted method</i>					
Size premium	+	–	–	–	
B/M premium	+	+	–	–	
E/P premium	+	+	+	+	
CF/P premium	+	+	+	–	
D/P premium	+	–	–	–	
Notes: ^b FSD, first-degree stochastic dominance; SSD, second-degree stochastic dominance; FSDR, first-degree stochastic dominance with a risk-free asset; SSDR, second-degree stochastic dominance with a risk-free asset. The FSDR efficient sets are not shown in the table as the results are the same as the FSD efficient sets. ^a As noted in Footnote 13, the range of r_f between 0.4 and 6.0 percent, is generally used as the testing interval. ^c The efficient set (winners) is marked by “+” and the inefficient set (losers) is marked by “–”					

As for the VW method, the E/P and CF/P premiums still outperform the other three premiums in the first sub-sample period and the B/M premium dominates the other four premiums in the second sub-sample period.

Generally speaking, the results of the two sub-sample periods for nonzero- and zero-investment strategies are almost the same as the findings of the whole sample period for the two methods.

4.2 Different holding periods

The dominant relationship of the one-quarter, two-year and three-year holding periods is also explored in order to check whether the results of different holding periods are the same as or similar to those of the one-year holding period in this subsection.

For the two methods, the Size 1, B/M 10, E/P 10, CF/P 10 and D/P 8 portfolios are generally still the winners for each indicator, respectively, for the one-quarter, two-year and three-year holding periods. These results are the similar to what we document for the one-year holding period.

The results of the winner strategies of various nonzero-investment strategies for the five indicators for the EW method are: E/P 10 and CF/P 10 strategies for one-quarter holding period and B/M 10 and CF/P 10 strategies for two- and three-year holding periods. Next, for the VW method, the winner strategies of various nonzero-investment strategies are: E/P 9 and CF/P 10 strategies for one-quarter holding period and E/P 10 and CF/P 10 strategies for two- and three-year holding periods.

According to the evidence, the winner strategies of the five zero-investment strategies for five indicators for the EW method are: B/M premium for one-quarter holding period and B/M and CF/P premiums for two- and three-year holding periods. Besides, for the VW method, the winner strategies of various zero-investment strategies are: CF/P premium for one-quarter holding period and E/P and CF/P premiums for two- and three-year holding periods.

The comparisons of the five zero-investment strategies present that the B/M and CF/P premiums dominate the other premiums for the EW method for three different holding periods. As for the VW method, the evidence shows that the E/P and CF/P premiums are the winners for these different holding periods.

4.3 Simulations

We simulate our trading strategies by annually investing \$100 to investigate whether there are more profits to be generated by our winner strategies and to check whether the EW method outperforms the VW method (see Figure 2).

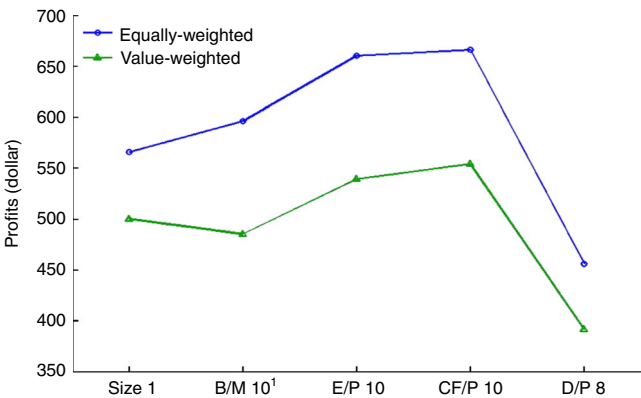
For example, the sum of annual profits or losses of the five nonzero-investment strategies for the first sub-sample period is calculated. From Figure 2, we first find that both the E/P 10 and CF/P 10 strategies outperform the other three strategies for the EW and VW methods, respectively. Second, the profits of each winner strategy for the EW method are apparently higher than those of each corresponding strategy for the VW method[21]. Specifically, the sums of profits of the five nonzero-investment strategies for the EW and VW methods are \$2,944 and \$2,470[22], respectively, which findings do support that EW portfolios outperform VW portfolios.

5. Significance tests and practical implications

While the results in Tables II-V indicate the presence of SD, the statistical significance of the results is not obvious. To provide the guidance about the likelihood that the

results in Tables II-V may be due to chance, randomization tests are undertaken. We use Seyhun's (1993) logic and procedure to execute the significance tests and examine the size portfolios of Table II, for example. First, from the entire sample of 570 annual returns for EW portfolios (57 years from 1952 to 2009 for the ten size portfolios, 57×10), two random samples of 57 returns are sampled with replacement. The two samples are arbitrarily labeled as "Size 1" and "Non-Size 1" returns and are examined for the presence of FSD, SSD, FSDR, and SSDR of the "Size 1" returns over the "Non-Size 1" returns, and the findings are recorded. Second, the entire procedure is repeated 1,000 times. We also apply the same procedure to test whether the "Size 1" returns dominate "Non-Size 1" returns for VW portfolios.

Table VI shows the findings of the randomization tests. It appears that observing SSDR for pairs of randomly drawn portfolios containing 57 annual observations is small. Out of 1,000 repetitions for EW portfolios, as shown in the first row, only 29 (2.9 percent) display SSDR. SSD is somewhat more common. Out of 1,000 repetitions, 40 display FSD (4.0 percent), 283 display SSD (28.3 percent) and 55 display FSDR (5.5 percent). Moreover, for the Size 1 portfolio to dominate 9 other size and the market portfolios simultaneously by SSDR due to chance will be even smaller to the extent that randomly chosen portfolios' returns will be less than perfectly correlated. Hence, the SSDR results obtained for the smallest size portfolio, the Size 1 (winner), cannot be attributed to chance[23].



Note: B/M 8 is the winner strategy for the VW methods in the first sub-sample period

Figure 2.
The sum of the profits or losses of the five nonzero-investment strategies in the first sub-sample period for the EW and VW methods

Table VI.
Frequency of stochastic dominance in 1,000 random replications

	FSD	SSD	FSDR	SSDR
Equally-weighted method	40	283	55	29
Value-weighted method	38	288	51	25

Notes: The annual data of the size portfolios are from 1952 to 2009. With annual data, two random vectors of 57 observations are sampled with replacement, and stochastic dominance is recorded. Then the procedure is repeated for 1,000 times

In short, the practical implications of our findings are discussed as follows. First, the SD findings suggest that investors or portfolio managers can allocate their funds between risky and risk-free assets to maximize their profits. That is, in the age of low interest rates for the last decade in the US treasury bill (bond) markets, the portfolio managers can invest part of the money in the US stocks and the remaining in the risk-free assets of non-US treasury bill (bond) markets to make more profits under the circumstances of global financial markets liberalization. Next, our simulation results again prove that the profits of each nonzero-investment or zero-investment strategy for EW portfolios are higher than those of each corresponding strategy for VW portfolios. The implication of these phenomena for investors is that the weighting of individual stocks within the portfolio is as important as the investment timing and selection decisions. Finally, the results also reveal that the investors or portfolio managers can invest in the highest E/P and/or CF/P portfolios (or their corresponding premiums) to make more profit or less loss.

6. Conclusions

Numerous previous studies employ the CAPM or market model to compare the relative performance among some trading strategies. This study first uses the SD theory to examine the relative performance of the five popular nonzero- and zero-investment strategies for the EW and VW methods, respectively.

Our main results show that both the highest E/P and CF/P strategies (and their corresponding premiums) generally produce higher returns than the other three strategies (and their corresponding premiums) for the EW and VW methods, respectively. The same or similar phenomena are found in each sub-sample period and different holding periods.

Finally, the results imply that the investors or the portfolio managers are recommended to invest in the highest E/P and/or CF/P portfolios (or their corresponding premiums) to make more profit or less loss for US stock markets.

Notes

1. Levy (1996) identifies four categories of anomalies: firm, accounting, seasonal and event anomalies. This study focusses on five popular firm and accounting anomalies for US stock markets.
2. Since investors must make investments with their own capital for the strategies, so we call them nonzero-investment or nonzero-cost strategies. As in previous studies, the winner portfolio of a certain indicator, such as Size or earnings-to-price (E/P) ratio, is dubbed as a strategy in this study.
3. Their findings show that the CF/P, B/M and S/P strategies offer the best performance in different industries.
4. As in footnote 2, the winner portfolio of a certain indicator is deemed a strategy.
5. Block and French (2002) document that funds tend to be EW to a greater degree than they are VW examining a sample of 506 actively managed mutual funds.
6. The meaning of EW (VW) portfolios is the same as that of the EW (VW) method, so the two terms are interchangeably used in this study.

7. In addition, the Jensen Measure, Treynor Measure and Sharpe Measure are often used to compare the performance among various strategies in many previous studies, such as Senchack and Martin (1987) and Park and Lee (2003).
8. As pointed out by Fama (1970), Levy and Lerman (1985) and Seyhun (1993), most tests of the above anomalies are a joint test of the CAPM validity and market efficiency. As a result, it is difficult to determine whether a rejection of joint hypothesis indicates market inefficiency or failure of the CAPM (or both).
9. The other advantage of the SD theory is that it is a pure test model, asserted by Levy and Lerman (1985), without adjusting for risk. This feature is meaningful to some investors as they are not familiar with the measurement of risk, such as the standard deviation or β .
10. Allocating investors' capital between the risky and risk-free assets can help distinguish the performance among various strategies, and utilizing US and non-US financial markets can help investors earn more profits in the age of the low risk-free interest rates. Empirical Results section discusses these issues in great detail.
11. This implies that the investment decision point of the Kenneth R. French Data Library is made in July of each year, after the publication of the annual financial statements.
12. The annual portfolio returns are calculated as $R = (1 + r_1)(1 + r_2) \dots (1 + r_{12}) - 1$, where R is annual portfolio returns and r_f ($i = 1, 2, 3, \dots, 12$) is portfolio returns on the i th month.
13. The readers interested in more details about the SD and SDR rules can consult Levy and Kroll (1979), Seyhun (1993), Elton and Gruber (1995), Best *et al.* (2000) and Post (2003). The SD theory given here is slightly modified from these papers.
14. The risk-free interest rate (r_f) of this study is taken from the Federal Reserve Bank of St. Louis. The minimum and maximum values of r_f are 0.4 and 14.7 percent and its mean and standard deviation are about 6.0 and 2.5 percent during the study period. However, the mean of the risk-free interest rate during the last decade is about 1.98 percent. For implementing well in practice in the age of the low risk-free interest rate, the range of r_f , between 0.4 and 6.0 percent, is generally used as the testing interval.
15. The meaning of the size in stochastic dominance theory is the number of winner portfolios or strategies in the efficient set.
16. As noted in Footnote 14, the mean value of r_f during the last decade is only about 1.98 percent which is far less than some of our higher testing levels of r_f , such as 6.1 percent. During the same period, however, the mean values of r_f for some countries, like New Zealand, Australia and India, are about 5.5, 5.0 and 7.7 percent, respectively. Under the global financial markets liberalization circumstances, the capital can move among the international markets. That is, the investors can invest part of their money in US stocks and the remaining money in other countries' risk-free assets (neglecting the exchange risks).
17. The comparative results within each indicator for the VW method are the same as those for the EW method. For space consideration, the results are not reported, but are available upon request.
18. The size (the number of the winner portfolios or strategies) of the FSDR or SSDR efficient set is a decreasing function of the r_f , only the B/M premium is included in the SSDR efficient set for $r_f \geq 4.4$ percent.
19. To save space, the detailed results are not shown but are available on request to interested readers.

20. The B/M 8 portfolio, instead of the B/M 10 portfolio, outperforms all the other B/M portfolios and the market during 1952/7-1980/6.
21. The results of the remaining simulations also present that our winner strategies can make more profits and support that the EW method outperforms the VW method.
22. For the EW method, the profits of the Size 1, B/M 10, E/P 10, CF/P 10 and D/P 8 strategies are \$566, \$596, \$666, \$660 and \$456, respectively, and the total profits of these winner strategies are \$2,944. For the VW method, the total profits of the Size 1, B/M 8, E/P 10, CF/P 10 and D/P 8 are \$500, \$485, \$539, \$554 and \$392, respectively, the total profits of these winner strategies are \$2,470.
23. In fact, we test a hypothesis that $H_0: p_{SSDR} = p_{NSSDR}$, $H_1: p_{SSDR} < p_{NSSDR}$. p_{SSDR} is the probability of the "Size 1" over the "Non-Size 1" by the SSDR rule and that p_{NSSDR} is the probability of the "Size 1" over the "Non-Size 1" by the non-SSDR rules. For EW portfolios of Table VI, $p_{SSDR} = 0.029$ (29/1,000) and $p_{NSSDR} = 0.378$ ((40+283+55)/1,000), the value of Z statistics is -19.38 . Since $-19.38 < -2.33$ (at the 1 percent significance level), we reject H_0 and conclude that the SSDR results obtained for the Size 1 (winner) cannot be attributed to chance. For VW portfolios, the value of Z statistics is -19.64 , which also rejects H_0 at the 1 percent significance level.

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Further reading

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